

O — Oceans

OCR B (Salters) H433 storyline **O**. The oceans and the carbon cycle: dissolving and the enthalpy changes that go with it, acid-base equilibria and pH, buffers, solubility products and entropy.

1. Born-Haber Cycles and Lattice Enthalpy

Lattice Enthalpy

Lattice enthalpy of formation ($\Delta_{LE}H^\circ$): The enthalpy change when one mole of an ionic compound is formed from its gaseous ions under standard conditions. This is exothermic (negative value).

Lattice enthalpy of dissociation: The enthalpy change when one mole of an ionic compound is completely separated into gaseous ions. This is endothermic (positive value) and is the reverse of lattice enthalpy of formation.

Lattice enthalpy cannot be measured directly. It is calculated indirectly using a Born-Haber cycle, which is an application of Hess's law applied to the formation of an ionic compound from its elements.

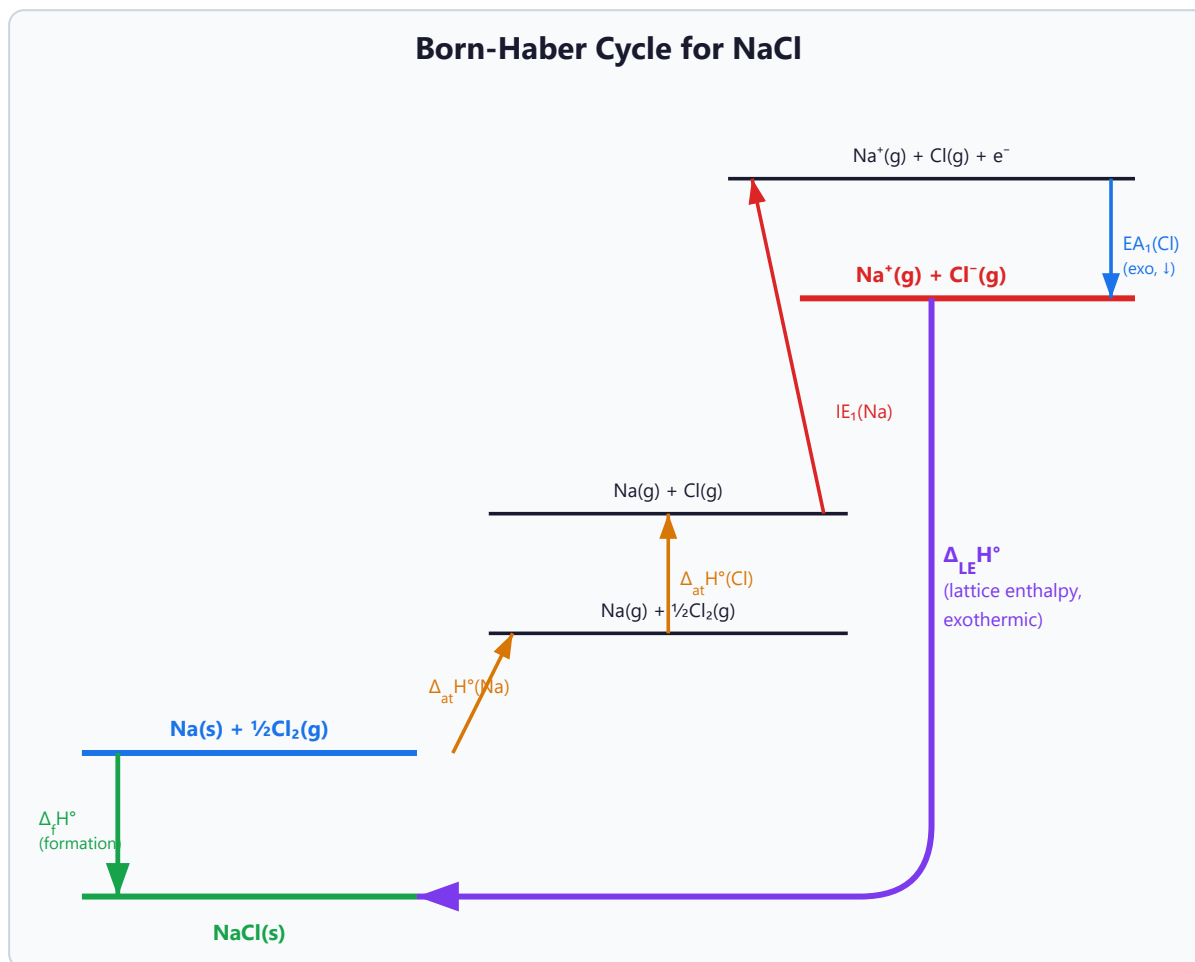
The Born-Haber Cycle

A Born-Haber cycle for a compound like NaCl includes the following steps:

1. Atomisation of sodium: $\text{Na(s)} \rightarrow \text{Na(g)} - \Delta_{\text{at}}H^\circ$
2. First ionisation energy of sodium: $\text{Na(g)} \rightarrow \text{Na}^+(\text{g}) + \text{e}^- - \text{IE}_1$
3. Atomisation of chlorine: $\frac{1}{2}\text{Cl}_2(\text{g}) \rightarrow \text{Cl(g)} - \Delta_{\text{at}}H^\circ$
4. First electron affinity of chlorine: $\text{Cl(g)} + \text{e}^- \rightarrow \text{Cl}^-(\text{g}) - \text{EA}_1$
5. Lattice enthalpy of formation: $\text{Na}^+(\text{g}) + \text{Cl}^-(\text{g}) \rightarrow \text{NaCl(s)} - \Delta_{LE}H^\circ$

By Hess's law: $\Delta_f H^\circ(\text{NaCl}) = \Delta_{\text{at}}H^\circ(\text{Na}) + \text{IE}_1(\text{Na}) + \Delta_{\text{at}}H^\circ(\text{Cl}) + \text{EA}_1(\text{Cl}) + \Delta_{LE}H^\circ$

Electron affinity: The enthalpy change when one mole of gaseous atoms gains one mole of electrons to form one mole of gaseous 1- ions. The first electron affinity is usually exothermic (negative); the second is always endothermic (positive) because energy is needed to overcome the repulsion between the incoming electron and the already negative ion.



Enthalpies of Hydration and Solution

Enthalpy of hydration ($\Delta_{\text{hyd}}H^\circ$): The enthalpy change when one mole of gaseous ions is dissolved in sufficient water to form an infinitely dilute solution. Always exothermic because ion-dipole bonds form between the ions and water molecules.

Enthalpy of solution ($\Delta_{\text{sol}}H^\circ$): The enthalpy change when one mole of an ionic compound dissolves in sufficient water to form an infinitely dilute solution.

$$\Delta_{\text{sol}}H^{\circ} = -\Delta_{\text{LE}}H^{\circ} + \Sigma\Delta_{\text{hyd}}H^{\circ}$$

A compound dissolves if the enthalpy of hydration of the ions is sufficiently exothermic to compensate for the energy needed to break apart the lattice. Smaller ions with higher charges have more exothermic hydration enthalpies because they create stronger electric fields that attract water molecules more strongly.

2. Entropy and Gibbs Free Energy

Entropy

Entropy (S): A measure of the disorder or the number of ways energy can be distributed among the particles in a system. Measured in $\text{J K}^{-1} \text{mol}^{-1}$. The greater the disorder, the greater the entropy.

General trends: gases have much higher entropy than liquids, which have higher entropy than solids. Entropy increases with temperature. Entropy increases when the number of moles of gas increases in a reaction.

$$\Delta S_{\text{system}} = \Sigma S^{\circ}(\text{products}) - \Sigma S^{\circ}(\text{reactants})$$

A positive ΔS indicates an increase in disorder (more ways of arranging energy). The Second Law of Thermodynamics states that for a spontaneous process, the total entropy of the universe must increase: $\Delta S_{\text{total}} > 0$.

Gibbs Free Energy

Gibbs free energy change (ΔG): A thermodynamic quantity that combines enthalpy and entropy to predict whether a reaction is feasible (thermodynamically spontaneous). A reaction is feasible when $\Delta G < 0$.

$$\Delta G = \Delta H - T\Delta S$$

where ΔG is in kJ mol^{-1} , ΔH is in kJ mol^{-1} , T is in K, and ΔS is in $\text{kJ K}^{-1} \text{mol}^{-1}$ (convert from J by dividing by 1000).

ΔH	ΔS	ΔG	Feasibility
Negative (exo)	Positive	Always negative	Feasible at all temperatures
Negative (exo)	Negative	Negative at low T	Feasible at low temperatures
Positive (endo)	Positive	Negative at high T	Feasible at high temperatures
Positive (endo)	Negative	Always positive	Never feasible

At the temperature where $\Delta G = 0$, the system is at equilibrium. Rearranging: $T = \Delta H / \Delta S$ gives the temperature at which the reaction becomes feasible.

Exam Tip: Feasibility ($\Delta G < 0$) does not mean the reaction will actually occur at an observable rate. Kinetic barriers (high activation energy) may prevent a thermodynamically feasible reaction from happening. For example, the combustion of diamond in air is thermodynamically feasible but kinetically extremely slow at room temperature.

3. Acids and Bases

Brønsted-Lowry Theory

Brønsted-Lowry acid: A proton (H^+) donor.

Brønsted-Lowry base: A proton (H^+) acceptor.

Conjugate acid-base pair: Two species that differ by a single proton. When an acid donates a proton, it forms its conjugate base. When a base accepts a proton, it forms its conjugate acid.

Example: $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$. Here HCl is the acid (donates H^+) and H_2O is the base (accepts H^+). Cl^- is the conjugate base of HCl, and H_3O^+ is the conjugate acid of H_2O .

Strong and Weak Acids

Strong acid: An acid that fully dissociates (ionises) in aqueous solution. Examples: HCl, H₂SO₄, HNO₃.

For a strong monobasic acid, $[H^+] = [\text{acid}]$.

Weak acid: An acid that only partially dissociates in aqueous solution. An equilibrium exists between the undissociated acid and its ions. Examples: CH₃COOH, H₂CO₃.

pH Calculations

$$\text{pH} = -\log_{10}[H^+]$$

$$[H^+] = 10^{-\text{pH}}$$

Strong acid pH: For a strong monobasic acid at concentration c , $[H^+] = c$, so $\text{pH} = -\log c$.

Strong base pH: For a strong base like NaOH at concentration c , $[OH^-] = c$. Use K_w to find $[H^+]$: $[H^+] = K_w / [OH^-]$, then calculate pH.

The Acid Dissociation Constant K_a

For a weak acid $HA \rightleftharpoons H^+ + A^-$:

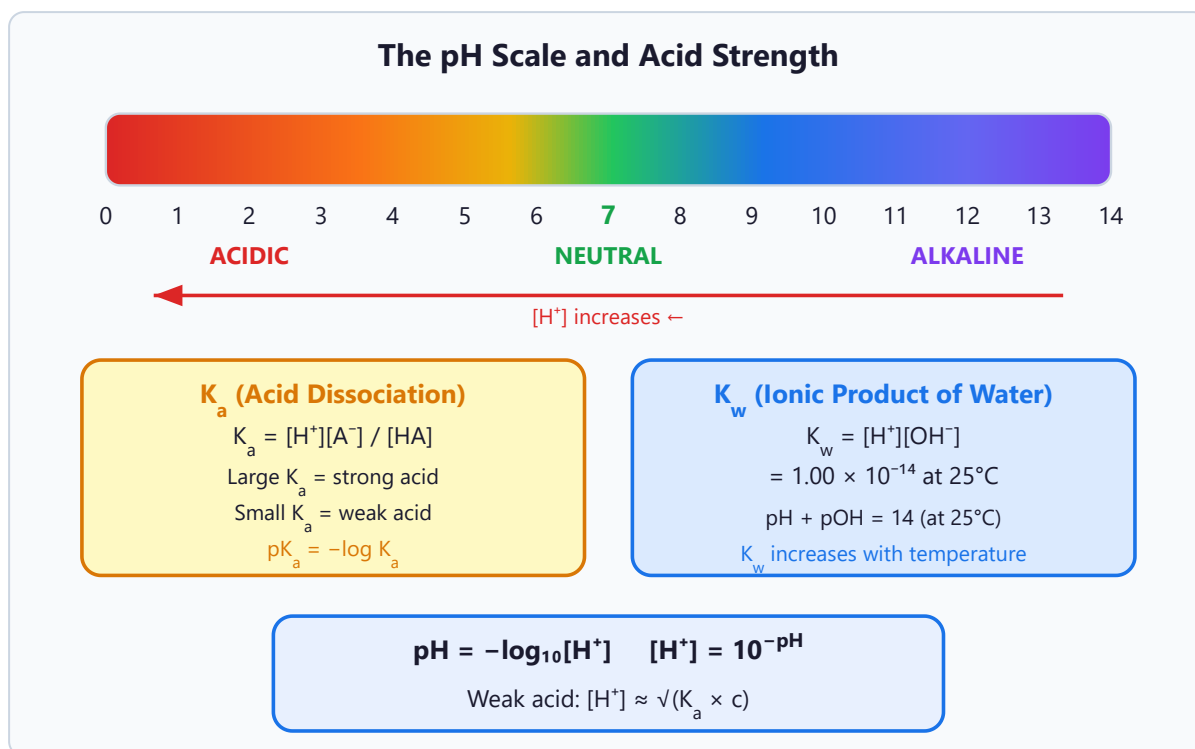
$$K_a = [H^+][A^-] / [HA]$$

For a weak acid, we assume $[H^+] = [A^-]$ (because the autoionisation of water contributes negligibly) and that the extent of dissociation is so small that the equilibrium concentration of HA is approximately equal to the initial concentration. Therefore:

$$[H^+] \approx \sqrt{(K_a \times c)}$$

$$\text{p}K_a = -\log_{10} K_a$$

The smaller the K_a (or larger the pK_a), the weaker the acid.



The Ionic Product of Water K_w

K_w : The ionic product of water. At 25 °C, $K_w = [H^+][OH^-] = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. K_w increases with temperature because the dissociation of water is endothermic.

In pure water at 25 °C: $[H^+] = [OH^-] = 1.00 \times 10^{-7} \text{ mol dm}^{-3}$, giving $pH = 7.00$. At higher temperatures, K_w is larger, so $[H^+]$ is greater and the pH of pure water is less than 7 — but the water is still neutral because $[H^+] = [OH^-]$.

4. Buffer Solutions

Buffer solution: A solution that resists changes in pH when small amounts of acid or base are added. It consists of a weak acid and its conjugate base (or a weak base and its conjugate acid) in roughly equal concentrations.

How Buffers Work

An acidic buffer (e.g. ethanoic acid + sodium ethanoate) contains a large reservoir of both HA molecules and A^- ions:

- **When acid is added:** The extra H^+ ions react with the conjugate base A^- to form HA. This removes the added H^+ , so the pH changes very little: $H^+ + A^- \rightarrow HA$
- **When base is added:** The extra OH^- ions react with the weak acid HA to form A^- and water. This removes the added OH^- : $OH^- + HA \rightarrow A^- + H_2O$

The buffer works because HA and A^- are present in large, roughly equal concentrations, so small additions of H^+ or OH^- cause only tiny changes to the ratio $[HA]/[A^-]$ and hence tiny changes in pH.

Calculating Buffer pH

Starting from $K_a = [H^+][A^-] / [HA]$, rearranging gives:

$$[H^+] = K_a \times [HA] / [A^-]$$

Then $pH = -\log[H^+]$. The Henderson-Hasselbalch equation provides an equivalent form:

$$pH = pK_a + \log_{10}([A^-] / [HA])$$

When $[HA] = [A^-]$, $pH = pK_a$. This is also the point at which the buffer has maximum buffering capacity.

When choosing a buffer for a particular pH, select a weak acid whose pK_a is close to the desired pH.

Exam Tip: In buffer calculations, you can use moles instead of concentrations for $[HA]$ and $[A^-]$ because they are in the same solution volume, so the volume cancels in the ratio. Be careful when the buffer is made by adding a strong base to an excess of weak acid — work out the moles of HA remaining and A^- formed.

Worked Example — pH of a Weak Acid:

Calculate the pH of $0.100 \text{ mol dm}^{-3}$ ethanoic acid ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$).

Assumptions: $[H^+] = [A^-]$; $[HA]_{eq} \approx [HA]_{initial}$

$$[H^+] = \sqrt{K_a \times [HA]} = \sqrt{(1.74 \times 10^{-5} \times 0.100)} = \sqrt{(1.74 \times 10^{-6})} = 1.32 \times 10^{-3} \text{ mol dm}^{-3}$$

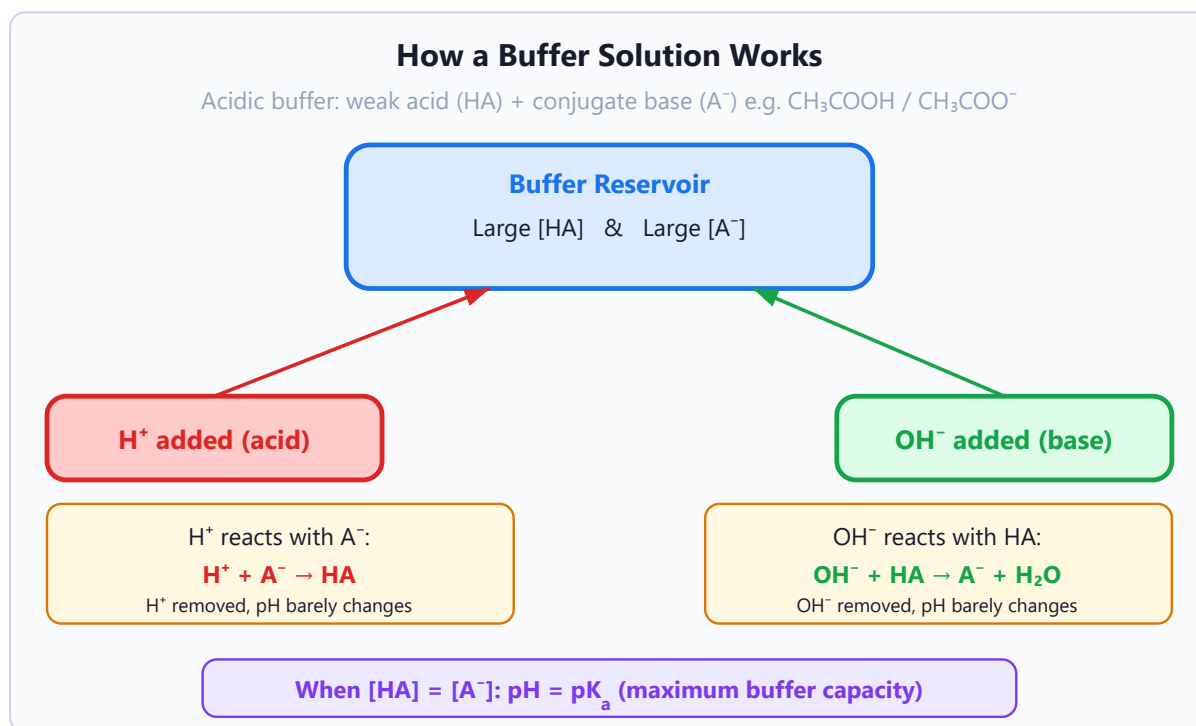
$$\text{pH} = -\log(1.32 \times 10^{-3}) = \mathbf{2.88}$$

Worked Example — Buffer pH:

A buffer is made from 0.200 mol CH_3COOH and 0.100 mol CH_3COONa . $K_a = 1.74 \times 10^{-5}$.

$$[H^+] = K_a \times [HA] / [A^-] = 1.74 \times 10^{-5} \times 0.200 / 0.100 = 3.48 \times 10^{-5} \text{ mol dm}^{-3}$$

$$\text{pH} = -\log(3.48 \times 10^{-5}) = \mathbf{4.46}$$



5. Titration Curves and Indicators

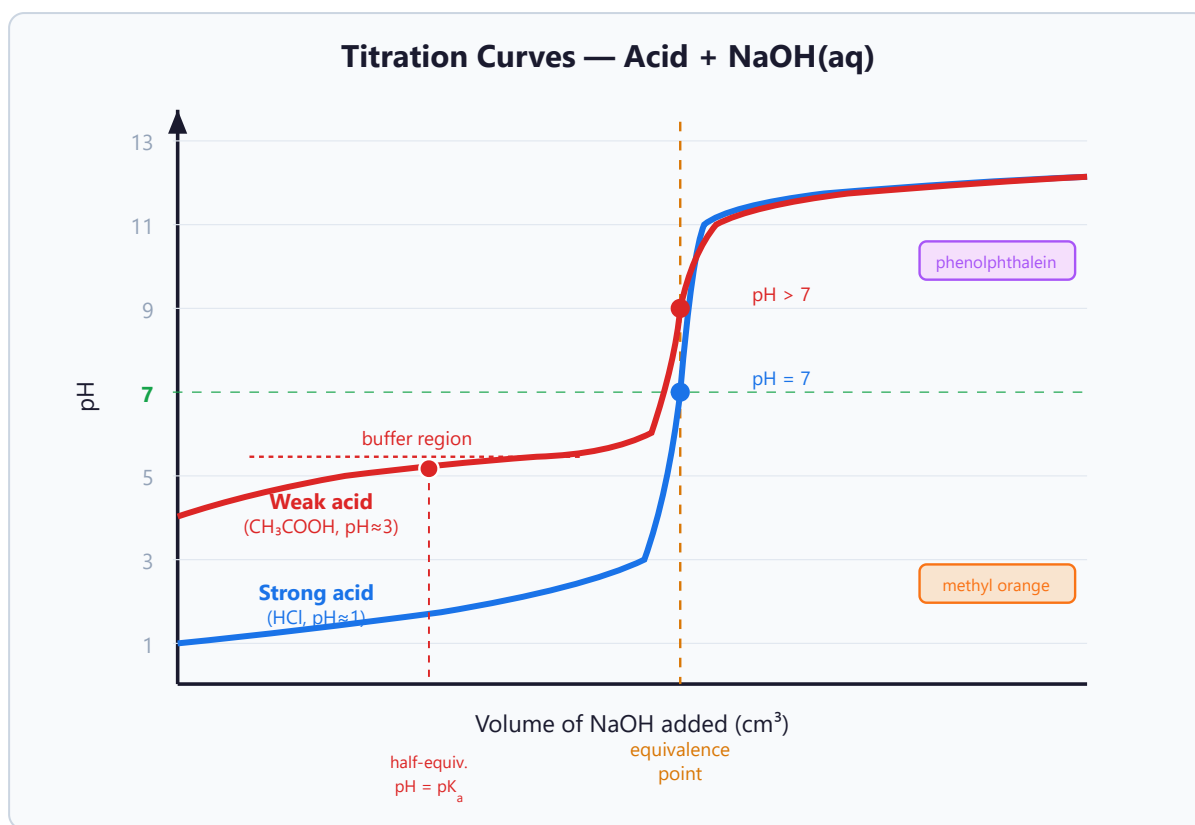
Titration Curves

A pH titration curve plots pH against volume of reagent added. The shape depends on the strengths of the acid and base:

Combination	pH at equivalence point	Suitable indicator
Strong acid + strong base	7	Any indicator with a range between pH 3–11 (e.g. phenolphthalein, methyl orange)
Strong acid + weak base	< 7 (acidic)	Methyl orange (range pH 3.1–4.4)
Weak acid + strong base	> 7 (basic)	Phenolphthalein (range pH 8.3–10.0)
Weak acid + weak base	$\approx$ 7 (variable)	No suitable indicator (gradual pH change, no sharp equivalence point)

Key features of a strong acid–strong base curve: the pH changes slowly initially, then very sharply near the equivalence point (a near-vertical section), then levels off. The equivalence point is at the centre of the vertical section.

For a weak acid–strong base titration: the initial pH is higher (because the acid is weak), the curve rises more gradually, and the equivalence point pH is above 7 because the salt formed is slightly basic (the conjugate base of the weak acid hydrolyses).



The Half-Equivalence Point

At the half-equivalence point (half the volume needed to reach the equivalence point), exactly half the weak acid has been neutralised, so $[HA] = [A^-]$. At this point, $pH = pK_a$. This provides an experimental method for determining K_a .

Indicators

Indicator: A weak acid (HIn) whose conjugate base (In^-) has a different colour. The indicator changes colour over a range of about 2 pH units centred on its pK_{In} value. A suitable indicator must have its colour change range within the steep (vertical) portion of the titration curve.

The indicator equilibrium: $HIn(aq) \rightleftharpoons H^+(aq) + In^-(aq)$. In acidic solution $[HIn]$ dominates (colour A); in basic solution $[In^-]$ dominates (colour B). The colour change occurs at $pH \approx pK_{In}$.

Neutralisation

Neutralisation is the reaction between an acid and a base to form a salt and water. For strong acid + strong base: $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$. The standard enthalpy of neutralisation for this reaction is

approximately $-57.1 \text{ kJ mol}^{-1}$. For reactions involving weak acids or bases, the enthalpy of neutralisation is less exothermic because some energy is used to complete the dissociation of the weak acid or base.

Exam Tip: When sketching titration curves, label the axes clearly (pH on y-axis, volume of titrant on x-axis). Mark the equivalence point, the initial pH, and the buffer region (the relatively flat section before the equivalence point in a weak acid–strong base titration). The buffer region is where pH changes least per unit volume of titrant added.